

motionbids: A lightweight Python package for BIDS Motion data curation

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Software

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Summary

motionbids is a Python package designed to facilitate the standardization of motion data according to the Brain Imaging Data Structure (BIDS). While BIDS was originally developed for neuroimaging (magnetic resonance imaging, MRI) (K. J. Gorgolewski et al., 2016), it has recently been extended to include motion data (Motion-BIDS) to support the growing field of behavioral neuroscience and Mobile Brain/Body Imaging (MoBI) (Jeung et al., 2024).

The package provides a lightweight, schema-driven interface for converting raw motion data, such as time series from optical motion capture systems or inertial measurement units (IMUs), into BIDS-compliant datasets including all relevant metadata. It aims to simplify the complexity of the BIDS file hierarchy and metadata requirements, allowing researchers to programmatically define channel attributes and export valid .tsv and .json sidecar files with minimal overhead.

Statement of Need

The recording of body movements is increasingly central to neuroscientific research, particularly in MoBI experiments where participants move freely while their brain activity is recorded using mobile electroencephalography (EEG) or functional near-infrared spectroscopy (fNIRS). In these multimodal setups, precise synchronization and data organization are important for subsequent analysis. However, unlike neuroimaging data, which has benefited from mature standardization protocols, motion data has historically lacked a unified format (Welzel et al., 2024). While the C3D format is widely used in the biomechanics community, its usage is mostly limited to marker-based motion capture systems. On the other hand, more versatile data formats such as EDF or XDF lack metadata standards appropriate for representing motion data. This fragmentation has made data sharing and reproducibility difficult, as datasets often rely on proprietary formats or ad-hoc directory structures.

The theoretical framework for standardizing this data was recently formalized in the "Motion-BIDS" extension proposal (BEP029), described by Jeung et al. (2024). While this publication established *how* motion data should be organized, the scientific community still lacks a dedicated, standalone tool to implement this standard practically.

motionbids addresses this need by providing a dedicated, standalone Python tool for converting generic motion data to the BIDS standard. It is designed for researchers who need to curate motion datasets for publication or archiving. By ensuring that datasets are strictly compliant with the Motion-BIDS schema, motionbids promotes the Findability, Accessibility,

41 Interoperability, and Reusability (FAIR) (Wilkinson et al., 2016) of behavioral data as well as
42 respecting the current BIDS version.

43 Multimodal Integration and Synchronization

44 A key feature of the Motion-BIDS extension—and a primary focus of `motionbids`—is its ability
45 to facilitate multimodal integration. In complex experimental designs involving simultaneous
46 recordings (e.g., EEG + Motion + Eye Tracking), data streams often have different sampling
47 rates and start times.

48 `motionbids` supports the generation of the `scans.tsv` file, a critical component for multimodal
49 alignment in BIDS. As specified in the BIDS standard, this file records the acquisition
50 time (`acq_time`, an ISO 8601 timestamp) of each data file. By accurately documenting
51 these timestamps, `motionbids` enables downstream analysis tools to temporally align motion
52 trajectories with neural time series, effectively bridging the gap between behavioral kinematics
53 and neural dynamics. This capability makes `motionbids` an essential utility for the MoBI
54 community, ensuring that motion data can be seamlessly integrated into broader neuroimaging
55 datasets.

56 State of the Field

57 The Python ecosystem for neuroimaging data management is robust, with widely cited packages
58 such as `NiBabel` (Brett et al., 2020) for file I/O, `Nipype` (K. Gorgolewski et al., 2011) for
59 pipeline orchestration, and `PyBIDS` (Yarkoni et al., 2019) for querying BIDS datasets.

60 However, a gap exists for the specific task of *creating* Motion-BIDS datasets: 1. **Lack of**
61 **Standalone Converters:** Currently, there is no dedicated package in Python or MATLAB
62 solely for converting motion data to BIDS. Existing BIDS conversion tools are embedded
63 within larger, modality-specific suites (e.g., MNE-BIDS (Appelhoff et al., 2019) or `FieldTrip`
64 (Oostenveld et al., 2011)). While `FieldTrip`'s `data2bids` can write Motion-BIDS compatible
65 files, robust conversion of motion data constitutes a substantial effort involving domain-
66 specific information, making it impractical to expect maintainers of M/EEG toolboxes to fully
67 support and sustain evolving motion-specific extensions. 2. **Complexity of Existing Tools:**
68 General-purpose converters like `HeuDiConv` (Halchenko et al., 2024) are primarily optimized for
69 DICOM-to-NIfTI conversion in MRI workflows. Adapting them for motion data often requires
70 significant “hacking” or manual intervention, which increases the risk of error and reduces
71 accessibility for behavioral researchers who may not be familiar with MRI workflows. 3. **Schema**
72 **Compliance:** `motionbids` is explicitly “schema-driven.” Unlike ad-hoc scripts, it validates user
73 input against the BIDS schema definitions for motion data, ensuring that required metadata
74 fields (e.g., `SamplingFrequency`, `TrackedPointsCount`) are present and correctly formatted
75 before the data is written to disk.

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78 AI Usage Disclosure

79 Generative AI tools were used during the development of `motionbids` and the preparation
80 of this paper. Specifically, Anthropic's Claude (Claude Opus 4.x models) was used to help
81 draft and refactor example conversion scripts and their documentation, copy-edit and structure
82 portions of this manuscript and cross-check bibliographic entries. The core software design,
83 the data model, and all scientific claims were conceived and decided by the human authors.

84 All AI-assisted output—code, documentation, and text—was reviewed, edited, and validated
85 by the authors, who take full responsibility for the content of the software and this paper.

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